



Traffic Stream Models on Road Section in Samarinda City (Case Study on AW. Syahrane Street)

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ABSTRACT

The interaction of loads in the transportation system is called traffic flow. Traffic flow interactions are used in transportation facility planning, design, and operation. This quantitative study focuses on Jalan Abdul Wahab Syahrane in Samarinda City. It employs the Greenshield method, which analyzes numerical data generated by the linear equation method, to find a mathematical relationship between vehicle speed and density, volume and density, and volume and velocity. According to the analysis findings, the higher the vehicle volume, the slower the vehicle speed, and the higher the density value

INTRODUCTION

Humans as users, vehicles as tools, and roads are critical to the movements. They interact with one another toward vehicles that meet the legislation's requirements and eligibility rules. The government then creates a level of road service with good conditions, whose capacity is directly proportional to the community's smooth level of mobility. The road can simplify or shorten the distance humans need to move places. The faster people move, the higher the level of development and economy in the region.

The volume of the traffic balance system of all types of roads is based on their function and hierarchy. It is critical in the service system. According to Adolf D. May (1990), traffic flow or volume describes the load in the transportation system. Traffic flow interactions are used in transportation facility planning, design, and operation.

The Greenshield method was used to determine a mathematical relationship between the speed and density, volume and density, and volume and velocity flow of vehicles on the road on Jalan Abdul Wahab Syahrani in Samarinda city. The location was chosen because no research had been conducted on it. A secondary arterial road connects the Samarinda Ulu District to the North Samarinda District. This road section is critical because many heavy-loaded vehicles pass through it as they enter or exit the regencies/cities in East Kalimantan's northern province.

LITERATURE REVIEW

There are three traffic flow characteristics, which are main parameters that are mathematically related to one another: 1) traffic flow (V) is the number of vehicles that pass through a specific point on a specific road segment in a given unit of time (vehicles/hour); 2) traffic density (D) is the number of vehicles in one unit of a given road length (vehicles/km); and 3) traffic speed (S) is the distance that can be covered in a given time unit (km/hour). Equation (1) expresses the mathematical relationship between speed, flow, and density.

$$V = D * S \quad (1)$$

Figure 1 depicts the general form of the mathematical relationship between Speed and Density (S-D), Flow and Density (V-D), and Flow-Speed (S-D) (V-S). Speed-Density has a downward monotony mathematical relationship, meaning speed decreases as traffic density increases. If the density is so high that vehicles cannot move again, traffic flow will be 0 (zero). It is a total traffic jam ($D = D_j$). No cars are on the road when the density is zero ($D = 0$), so the traffic flow is zero (zero). As a result, the behaviour of traffic flows that fall between these two extreme values must be investigated.

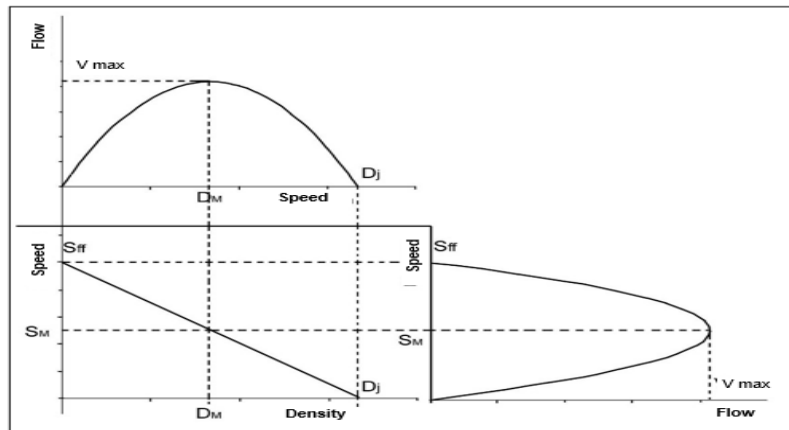


Figure 1. The Mathematical Relationship Between Velocity, Flow, and Density

The speed and traffic flow will decrease as the density increases from zero. If the density continues to rise, a point will be reached where the increase in density will not increase traffic flow but decrease it (Figure 1). Flow capacity denotes the maximum end of traffic flow. Figure 1 also depicts some important traffic flow parameters, which are as follows:

VM: maximum capacity or flow (vehicles/hour)

SM: the speed at top traffic flow conditions (km/h)

DM: density at full traffic flow conditions (vehicles/km)

Dj: Density in conditions of total traffic jam (vehicles/km)

Sff: the speed at very low traffic flow conditions or density conditions close to 0 (zero) or free flow speed (km/hour).

Because this condition occurs when no vehicles ($D = 0$), the free flow speed (S_{ff}) cannot be observed in the field. The value of free flow velocity can be calculated mathematically using the Flow-Speed relationship in the field. The flow and speed of traffic can be collected in the area by conducting a traffic survey. Because different types of traffic pass through, traffic flow data must be expressed in a single unit: the passenger car unit.

The Greenshields model is a general model discussed so far that relates speed, flow, and density. It is a linear model proposed by Greenshields that can measure speed, flow, and density. Greenshield concluded that the density relationship is a linear curve, according to Tamin (2000). Because the relationship function is the simplest and thus easy to apply, this linear relationship of speed and density has become popular in reviewing traffic flow movements. According to the Greenshield model, the relationship between the space mean speed and density in a traffic flow is linear. The Greenshield model employs a linear approach. According to Greenshield, the mathematical relationship between velocity and density is assumed to be linear (Tamin, 2000). Some researchers began to create traffic flow models based on curve matching and statistical tests. There are two methods for evaluating models: 1) The V - D - S relationship is tested for accuracy with actual field data; 2) the relationship is thought to meet certain boundary conditions: 2a) When the density is zero, the flow is zero; 2b) When the density is maximum, the flow is zero; 2c) When the density is zero, the mean free velocity occurs; and 2d) The flow-density curves are convex (in other

words, there is a point of maximum flow). The Greenshield model generates slopes and intersections by plotting straight lines or using linear regression. It states that the mathematical relationship Velocity-Density is assumed to be linear, as indicated by equation (2).

$$S = S_{ff} - \frac{S_{ff}}{D_j} \cdot D \quad (2)$$

Furthermore, the mathematical relationship between Flow-Density can be derived using the basic equation (1), and then by plugging equation (3) into equation (2), equations (4) and (5) can be derived;

$$S = \frac{V}{D} \quad (3)$$

$$\frac{V}{D} = S_{ff} - \frac{S_{ff}}{D_j} \cdot D \quad (4)$$

$$V = D \cdot S_{ff} - \frac{S_{ff}}{D_j} \cdot D^2 \quad (5)$$

Equation (5) is an equation that expresses the mathematical relationship between Flow-Density. The maximum flow condition (VM) can be obtained at the time when the flow $D = D_M$. The value of $D = D_M$ can be obtained through equations (6) and (7);

$$\frac{\partial V}{\partial D} = S_{ff} - \frac{2 \cdot S_{ff}}{D_j} \cdot D_M = 0 \quad (6)$$

$$D_M = \frac{D_j}{2} \quad (7)$$

By entering equation (7) into equation (5), then the value of VM can be obtained as seen in equation (8);

$$V_M = \frac{D_j \cdot S_{ff}}{4} \quad (8)$$

Furthermore, the mathematical relationship between Flow-Speed can be derived using the basic equation (1), and by plugging equation (9) into equation (2), it can be derived through equations (10), (11), and (12);

$$D = \frac{V}{S} \quad (9)$$

$$S = S_{ff} - \frac{S_{ff}}{D_j} \cdot \frac{V}{S} \quad (10)$$

$$\frac{S_{ff}}{D_j} \cdot \frac{V}{S} = S_{ff} - S \quad (11)$$

$$V = D_j \cdot S - \frac{D_j}{S_{ff}} \cdot S^2 \quad (12)$$

Equation (12) is an equation that expresses the mathematical relationship between Flow-Velocity. The maximum flow condition (VM) can be obtained when the flow is $S = S_M$. The value of $S = S_M$ can be obtained through equations (13) and (14);

$$\frac{\partial V}{\partial S} = D_j - \frac{2 \cdot D_j}{S_{ff}} \cdot S_M = 0 \quad (13)$$

$$S_M = \frac{S_{ff}}{2} \quad (14)$$

By entering equation (14) into equation (12), then the value of VM can be obtained as seen in the following equation (15);

So it can be concluded that VM can be achieved under conditions $S = S_M$ and $D = D_M$

$$V_M = \frac{D_j \cdot S_{ff}}{4} \quad (15)$$

By performing a linear transformation, equation (2) can be simplified and rewritten as the linear equation $Y_i = A + BX_i$ by assuming $S = Y_i$ and $D = X_i$. By knowing several sets of S_i and D_i data that can be obtained from the results of the survey of the speed and density of traffic flow, then using linear-regression analysis (equations 17 and 18), parameters A and B can be calculated and generated the following values;

$$A = S_{ff} \text{ dan } B = -\frac{S_{ff}}{D_j} \quad (16)$$

So, finally, we get the value of $S_{ff} = A$ and value $D_j = -\frac{A}{B}$.

$$B = \frac{N \sum_{i=1}^N (X_i - \bar{X})(Y_i - \bar{Y})}{N \sum_{i=1}^N (X_i - \bar{X})^2} \quad (17)$$

$$A = \bar{Y} - B\bar{X} \quad (18)$$

METHODOLOGY

The research method employs quantitative research, which focuses on the statistical analysis of numerical data. The subjects studied, data collected, data sources required, and data collection tools used all went as planned. Data is gathered through measurements using objective and standardized tools, which entails calculating numbers or quantifying data (Sudaryana et al., 2022).

Location and Object of Research

The study occurred at Jalan Abdul Wahab Syahrani in Samarinda (figure 2). Motorcycles (MC), light vehicles (LV), heavy vehicles (HV), and unmotorized vehicles are the vehicles being surveyed (UM).

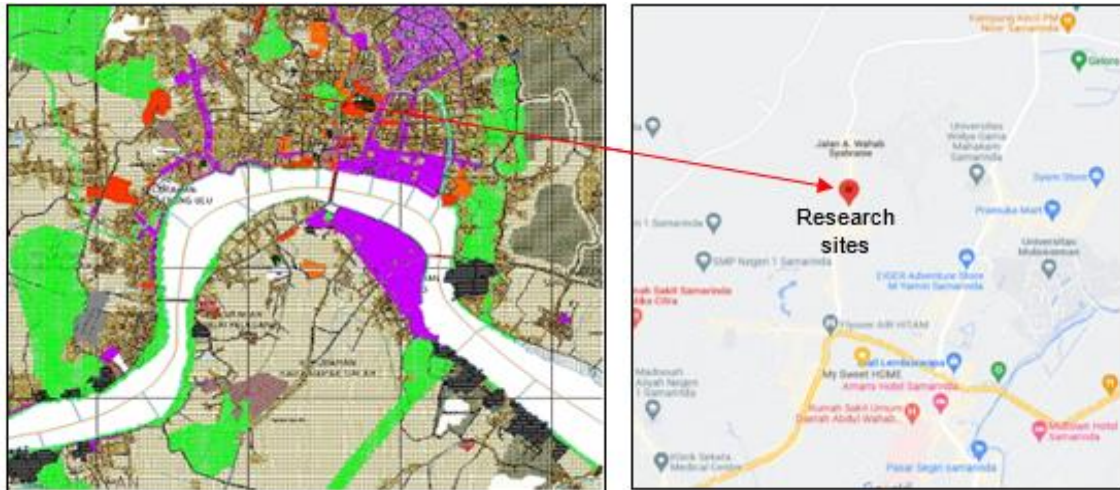


Figure 2. Research Sites

Preliminary Survey

A preliminary data collection or pilot survey is a small-scale survey conducted before the primary survey. Its goal is to inspect the survey location and the surrounding environment. Simultaneously, it verifies the level of conformity of the survey method and the compilation method used following the form format that will be used.

Data Collection

The goal of primary data collection is to collect field data for analysis. The survey was conducted to calculate the volume of all types of vehicles (table 1) using a hand counter and a survey form. A travel time survey is calculated by recording the trip's travel time along with the distance based on the observation distance. The surveyor then documents what is seen on the stopwatch on the survey form. Figure 3 depicts Jalan Abdul Wahab Syahrani's geometric data, which shows a road width of 12 meters, two lanes, and two directions without barriers. This road section's surroundings include several residential and commercial areas.

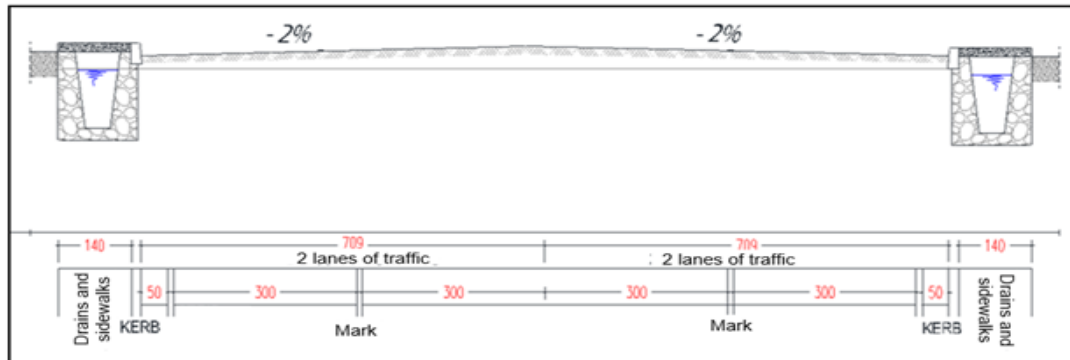


Figure 3. Road Segment Geometry Data

Table 1. Recapitulation of Vehicle Volume and Speed Data

Day	Time		Direction 1		Direction 2	
			Vehicle volume	Vehicle Speed	Vehicle volume	Vehicle Speed
			pcu/h	km/h	pcu/h	km/h
1	07.30	08.30	2,211	49.633	1,602	44.430
	08.30	09.30	2,052	45.215	1,631	42.956
	11.30	12.30	1,468	40.021	1,811	42.083
	12.30	13.30	1,684	38.853	1,604	42.530
	15.30	16.30	1,621	40.026	1,522	43.862
	16.30	17.30	1,973	38.780	2,026	42.771
2	07.30	08.30	2,379	43.341	1,759	45.783
	08.30	09.30	1,838	42.811	1,840	46.322
	11.30	12.30	1,715	40.610	1,819	43.702
	12.30	13.30	1,667	40.603	1,765	44.188
	15.30	16.30	1,568	43.331	1,730	44.857
	16.30	17.30	1,577	40.312	1,364	43.743
3	07.30	08.30	1,618	44.997	1,831	42.334
	08.30	09.30	1,628	44.880	1,480	40.934
	11.30	12.30	1,656	41.152	1,834	40.867
	12.30	13.30	1,701	43.304	1,916	38.448
	15.30	16.30	1,491	45.441	1,802	39.753
	16.30	17.30	1,539	43.649	1,461	39.879
4	07.30	08.30	1,549	45.893	1,841	44.402
	08.30	09.30	1,517	43.976	1,583	44.023
	11.30	12.30	1,540	43.124	1,790	40.610
	12.30	13.30	1,778	43.741	1,826	40.654
	15.30	16.30	1,661	43.915	1,724	41.326
	16.30	17.30	1,870	45.794	1,416	39.687

Graph of the vehicle volume data set per hour, each day in directions 1 and 2 (figures 4 and 5).

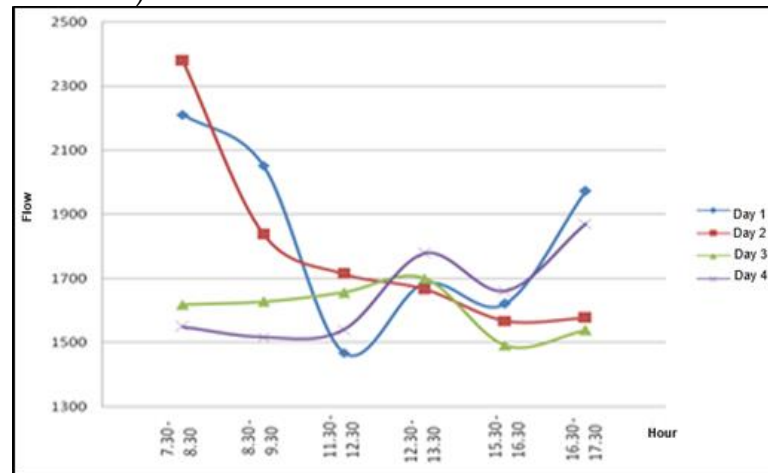


Figure 4. Vehicle Volume Data Direction 1

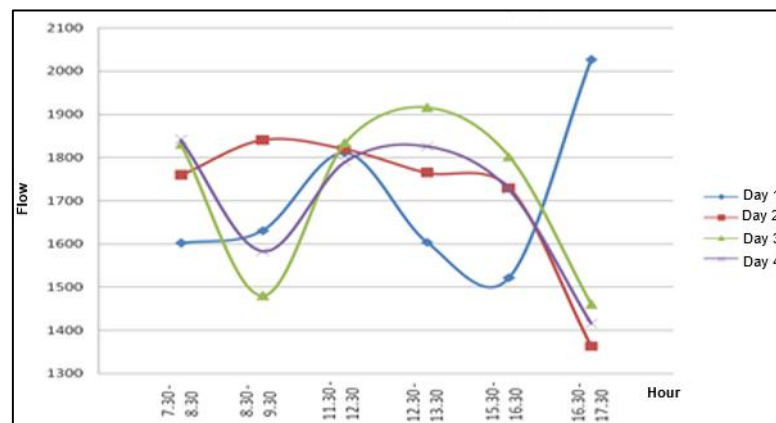


Figure 5. Vehicle Volume Data Direction 2

Graph of vehicle volume data sets per hour each day in directions 1 and 2 (figures 6 and 7).

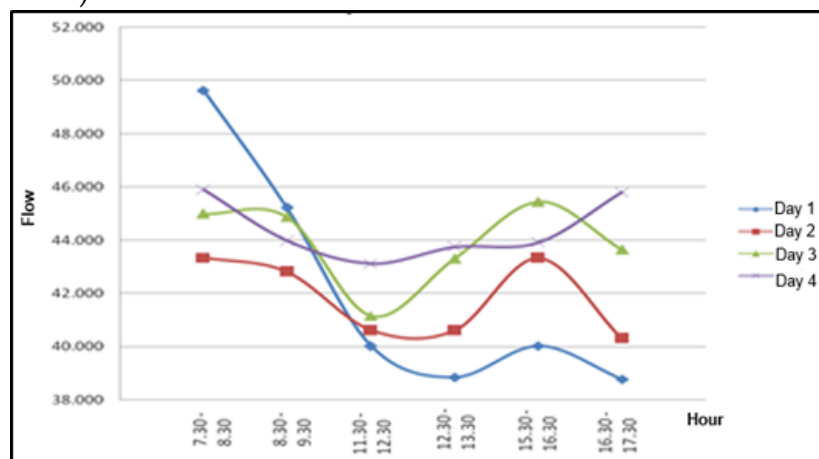


Figure 6. Vehicle Speed Data Direction 1

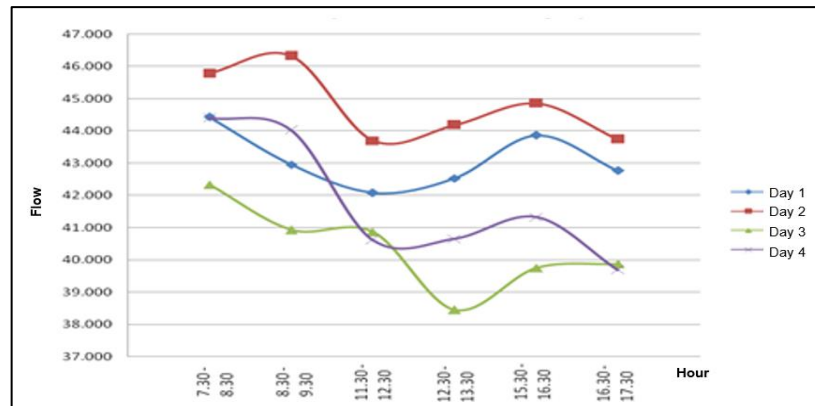


Figure 7. Vehicle Speed Data Direction 2

RESULTS AND DISCUSSION

At this stage, researchers use the literature analysis stage and Microsoft Excel software, as well as correlations and equations, to model the relationship between traffic characteristics of the Greenshield model. Table 2 shows the results of the Greenshield model analysis for Jalan Abdul Wahab Syahrane, Samarinda city, with 4 days of data acquisition for directions 1 and 2.

Table 2. Recapitulation of Greenshield Model Analysis Results

Hari	Clausal	Direction 1		Direction 2	
		Conditions (1)	Conditions (2)	Conditions (1)	Conditions (2)
1	$S - D$	$S = 50,548 - 0,193 D$		$S = 47,358 - 0,107 D$	
	$V - D$	$V = 50,548 D - 0,193 D^2$		$V = 47,358 D - 0,107 D^2$	
	$V - S$	$V = 261,909 S - 5,181 S^2$		$V = 442,601 S - 9,345 S^2$	
	D (pcu/h)	99,554	162,352	172,018	270,579
	S (km/h)	31,332	19,220	28,949	18,413
2	$S - D$	$S = 42,352 - 0,012 D$		$S = 47,233 - 0,064 D$	
	$V - D$	$V = 42,352 D - 0,012 D^2$		$V = 47,233 D - 0,064 D^2$	
	$V - S$	$V = 3529,33 S - 83,333 S^2$		$V = 738,015 S - 15,625 S^2$	
	D (pcu/h)	1296,416	2232,916	302,242	435,773
	S (km/h)	26,795	15,557	27,889	19,343
3	$S - D$	$S = 44,126 - 0,006 D$		$S = 46,246 - 0,136 D$	
	$V - D$	$V = 44,126 D - 0,006 D^2$		$V = 46,246 D - 0,136 D^2$	
	$V - S$	$V = 354,297 S - 166,666 S^2$		$V = 340,037 S - 7,353 S^2$	
	D (pcu/h)	2874	4480,333	105,364	234,680
	S (km/h)	29,983	45,857	31,917	14,327
4	$S - D$	$S = 53,314 - 0,237 D$		$S = 44,238 - 0,060 D$	
	$V - D$	$V = 53,314 D - 0,237 D^2$		$V = 44,238 D - 0,060 D^2$	
	$V - S$	$V = 224,953 S - 4,219 S^2$		$V = 737,312 S - 16,667 S^2$	
	D (pcu/h)	66,468	158,485	292,73	444,567
	S (km/h)	37,561	15,757	26,673	17,564
D average (pcu/h)		1084,1095	1758,5215	218,0885	346,39975
S average (pcu/h)		31,41775	24,09775	28,857	17,41175

Figures 8 and 9 show the mathematical relationship between speed and density, volume and density, and volume with velocity for 4 days based on the results of the Greenshield model analysis for Jalan Abdul Wahab Syahrane, Samarinda city, for directions 1 and 2.

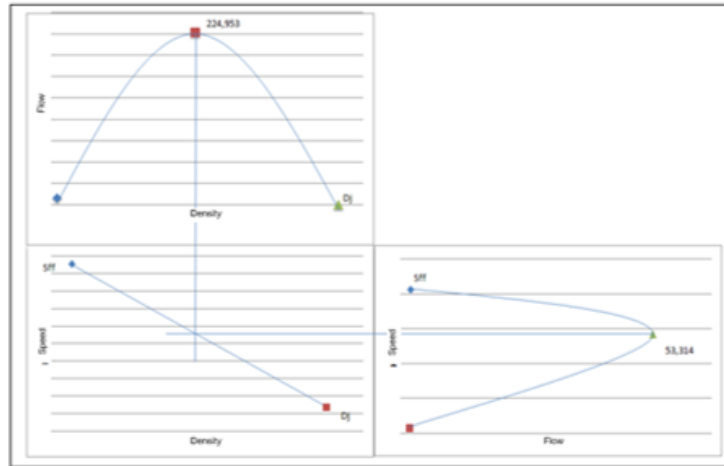


Figure 8. The Mathematical Relationship Between S - D, Volume - D, V - S
Direction 1

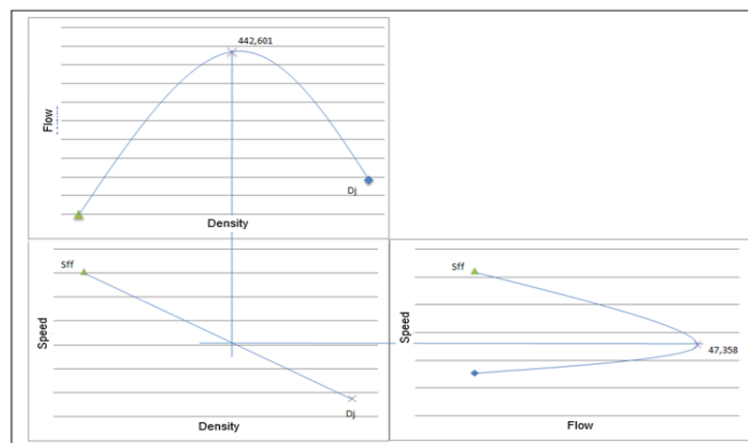


Figure 9. The Mathematical Relationship Between S - D, Volume - D, V - S
Direction 2

CONCLUSIONS AND RECOMMENDATIONS

Traffic flow characteristics of the Greenshield model on the Abdul Wahab Syahrane road, Samarinda City direction 1, condition 1 ; D average is 1084.1095 pcu/h, and S average is 31.41775 km/h, condition 2; D average is 1758.5215 pcu/h and S average = 24.09775 km/h. Direction 2, condition 1 ; D average is 218.0885 passenger car units/hour and S average = 28.857 km/h, condition 2; D average is 346.39975 passenger car units/hour and S = 17.41175 km/h on average. The results prove that the higher the volume of the vehicle, the slower the vehicle's speed and the higher the density value.

FURTHER STUDY

This research still has limitations, so it is necessary to carry out further research related to the topic of Traffic Stream Models on Road Section in order to improve this research and add insight to readers

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